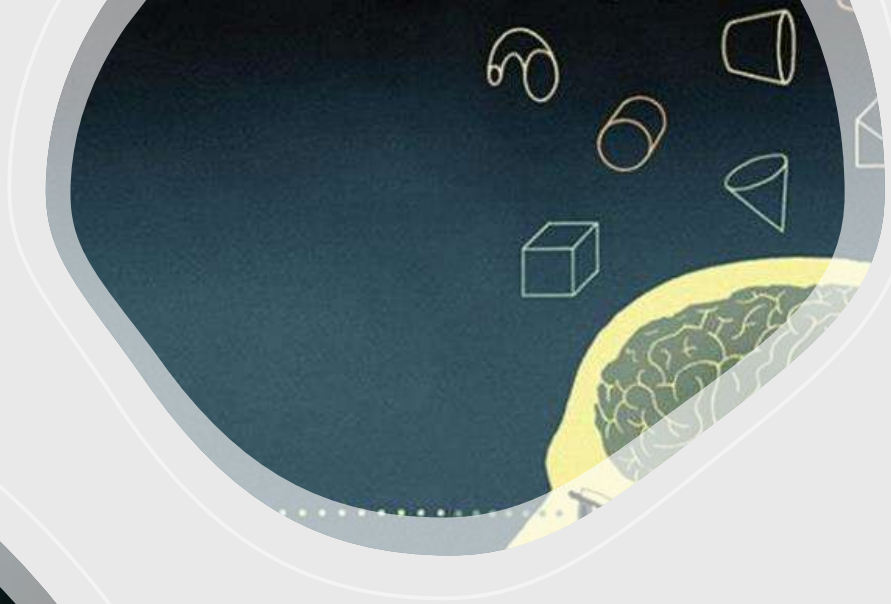
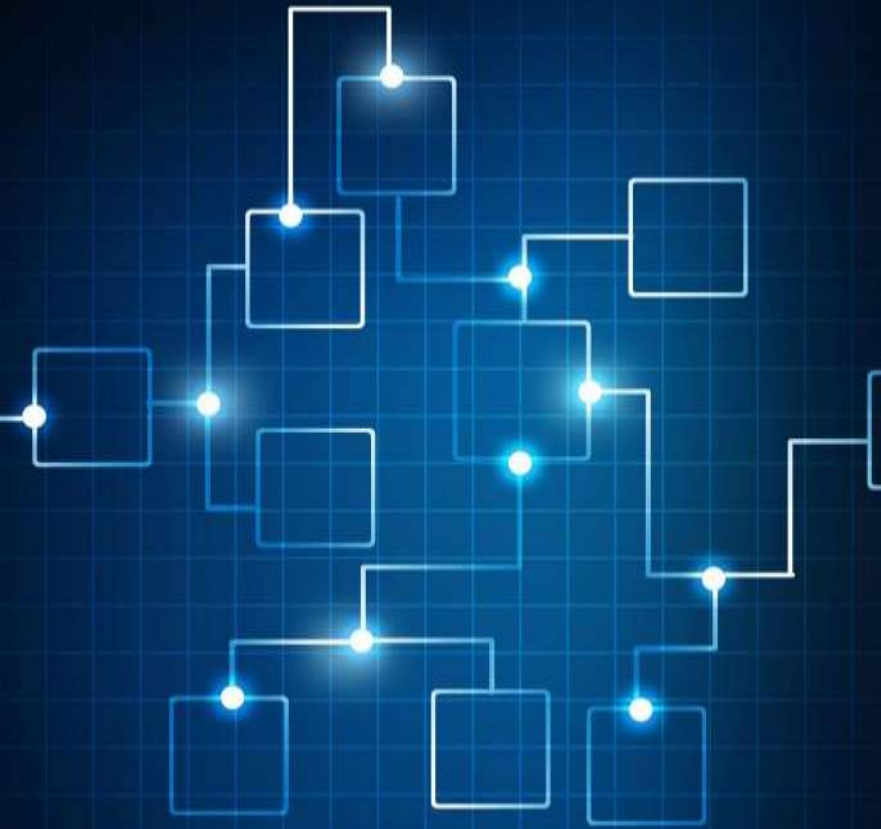




# Pattern recognition

An introduction

Dr. Ahmad S. Tarawneh



# Outline

- What is PR
- Classification
- Applications of PR
- PR system flowchart diagram



# Pattern recognition

- Humans are good at recognizing objects (or patterns)
  - It is generally easy to distinguish the sound of a human voice, from that of a violin
  - Handwritten numeral "3," from an "8"
  - We recognize faces around us
- Humans do it unconsciously
- Because we cannot explain our expertise:
  - We find it difficult to write a computer program to do the same



# Example

What is the content of  
this image?  
Is it easy to find out?



# Example

What about this?

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 157 | 159 | 159 | 104 | 104 | 115 | 128 | 131 | 133 | 133 | 132 | 131 | 132 | 130 | 129 | 118 | 132 | 158 | 156 | 153 | 190 | 144 | 117 | 126 | 120 | 81  |
| 159 | 165 | 153 | 101 | 103 | 113 | 126 | 129 | 130 | 130 | 126 | 124 | 127 | 128 | 127 | 120 | 122 | 158 | 159 | 154 | 160 | 190 | 121 | 118 | 67  | 47  |
| 162 | 154 | 154 | 98  | 101 | 114 | 124 | 127 | 130 | 132 | 144 | 159 | 155 | 132 | 123 | 119 | 119 | 148 | 154 | 150 | 140 | 185 | 161 | 60  | 48  | 45  |
| 141 | 132 | 158 | 93  | 98  | 110 | 121 | 125 | 122 | 129 | 143 | 172 | 191 | 188 | 143 | 105 | 117 | 148 | 140 | 145 | 142 | 153 | 105 | 44  | 49  | 71  |
| 100 | 130 | 157 | 93  | 99  | 110 | 120 | 116 | 116 | 129 | 138 | 163 | 191 | 205 | 211 | 130 | 107 | 153 | 98  | 133 | 147 | 107 | 44  | 47  | 81  | 151 |
| 87  | 130 | 157 | 92  | 97  | 109 | 124 | 111 | 123 | 134 | 139 | 175 | 194 | 201 | 207 | 205 | 126 | 151 | 74  | 114 | 160 | 57  | 49  | 63  | 141 | 163 |
| 93  | 131 | 159 | 92  | 98  | 112 | 132 | 108 | 123 | 133 | 162 | 180 | 183 | 192 | 196 | 205 | 184 | 151 | 138 | 199 | 195 | 54  | 47  | 119 | 161 | 156 |
| 96  | 134 | 164 | 95  | 97  | 113 | 147 | 108 | 125 | 142 | 156 | 171 | 173 | 178 | 184 | 181 | 186 | 191 | 206 | 203 | 161 | 44  | 84  | 158 | 159 | 155 |
| 95  | 137 | 165 | 95  | 95  | 111 | 168 | 122 | 130 | 137 | 145 | 139 | 144 | 139 | 145 | 179 | 193 | 203 | 194 | 158 | 95  | 49  | 135 | 160 | 157 | 155 |
| 101 | 139 | 166 | 94  | 96  | 104 | 172 | 130 | 126 | 130 | 108 | 77  | 85  | 80  | 153 | 191 | 188 | 161 | 144 | 113 | 48  | 83  | 161 | 160 | 156 | 153 |
| 101 | 133 | 167 | 94  | 96  | 100 | 154 | 137 | 123 | 92  | 67  | 57  | 72  | 153 | 182 | 184 | 175 | 101 | 116 | 53  | 48  | 119 | 166 | 163 | 159 | 152 |
| 99  | 130 | 169 | 97  | 99  | 109 | 131 | 128 | 84  | 55  | 60  | 75  | 149 | 176 | 170 | 194 | 209 | 99  | 79  | 51  | 67  | 150 | 158 | 155 | 154 | 151 |
| 97  | 129 | 170 | 97  | 98  | 118 | 122 | 94  | 66  | 56  | 56  | 140 | 161 | 114 | 136 | 187 | 163 | 81  | 85  | 52  | 98  | 161 | 159 | 154 | 148 | 137 |
| 92  | 123 | 173 | 101 | 98  | 129 | 95  | 74  | 74  | 45  | 94  | 174 | 106 | 115 | 126 | 168 | 108 | 60  | 92  | 55  | 128 | 157 | 153 | 148 | 145 | 157 |
| 81  | 115 | 175 | 104 | 116 | 87  | 78  | 69  | 84  | 56  | 140 | 124 | 158 | 170 | 143 | 173 | 150 | 76  | 90  | 68  | 148 | 153 | 146 | 148 | 186 | 196 |
| 69  | 108 | 172 | 107 | 103 | 87  | 82  | 54  | 83  | 105 | 93  | 107 | 153 | 166 | 132 | 162 | 153 | 68  | 87  | 97  | 157 | 149 | 141 | 179 | 204 | 206 |
| 71  | 119 | 172 | 106 | 91  | 78  | 97  | 70  | 99  | 104 | 59  | 116 | 142 | 153 | 141 | 165 | 123 | 55  | 84  | 132 | 154 | 146 | 148 | 199 | 209 | 210 |
| 61  | 126 | 175 | 112 | 83  | 74  | 92  | 123 | 130 | 53  | 61  | 108 | 137 | 132 | 138 | 154 | 77  | 58  | 82  | 150 | 152 | 143 | 155 | 210 | 211 | 213 |
| 53  | 128 | 175 | 105 | 71  | 82  | 109 | 127 | 75  | 50  | 57  | 74  | 115 | 139 | 151 | 117 | 47  | 67  | 89  | 154 | 154 | 143 | 159 | 218 | 214 | 199 |
| 56  | 115 | 173 | 105 | 61  | 76  | 106 | 114 | 70  | 54  | 52  | 60  | 102 | 137 | 160 | 146 | 78  | 67  | 96  | 135 | 130 | 125 | 165 | 215 | 142 | 81  |
| 117 | 106 | 176 | 101 | 55  | 71  | 81  | 112 | 101 | 57  | 55  | 70  | 117 | 139 | 152 | 188 | 198 | 112 | 87  | 146 | 131 | 112 | 178 | 164 | 81  | 91  |
| 107 | 121 | 177 | 89  | 50  | 64  | 60  | 103 | 114 | 66  | 56  | 90  | 120 | 140 | 149 | 169 | 201 | 194 | 100 | 148 | 134 | 155 | 208 | 120 | 99  | 99  |

# Patterns

• For example, each person's face is a pattern composed of a particular combination of structures.

- eyes, nose, mouth, etc., located in certain positions on the face
- Each of these are patterns that are unique for each person
- By analyzing sample images of faces, a program should be able to capture the **pattern** specific to a face and identify (or recognize) it as a face
  - as a member of a category or class we already know
- There may be several categories (or classes) and we have to sort (or classify) a particular face into a certain category (or class)
  - hence the term classification



# Recognition vs Classification

- These terms are closely related.
  - Recognition is a more general term and classification is a type of recognition
  - When the purpose is to sort patterns into a set of pre-defined classes, then recognition and classification have the same meaning
  - We can say classification is a step-in pattern recognition
- 
- 
- 
- 

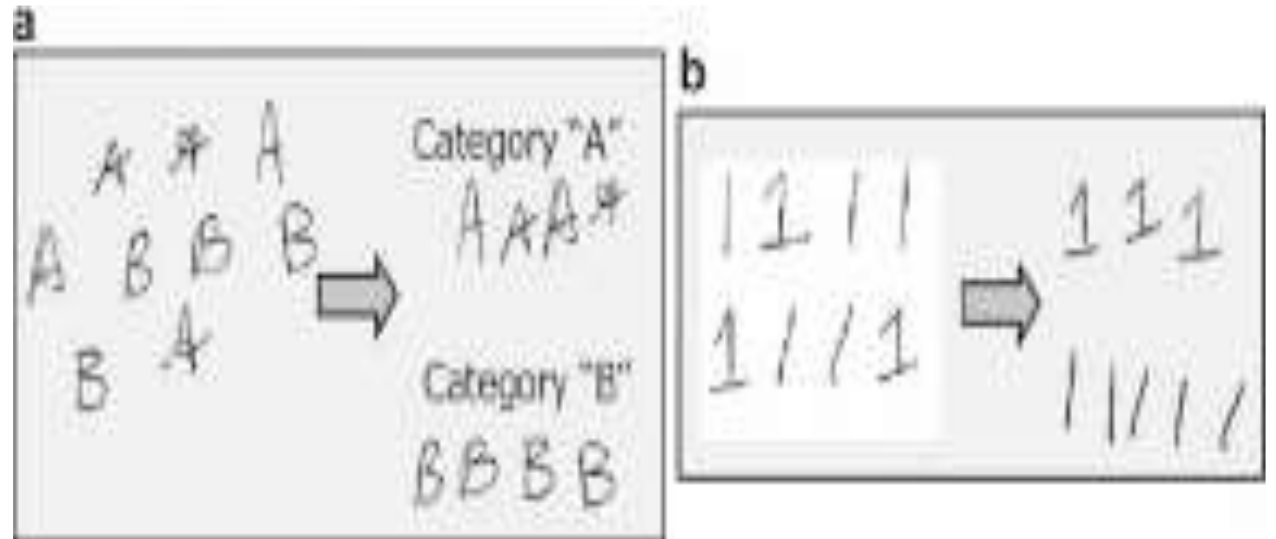
# Example

- Suppose you want to recognize a face in an image:
  - First you scan the image to find a face or several faces: (face detection)
    - some patterns distinguish faces from tables
    - This includes pattern recognition
  - There might be several objects in the image, the face is one of them
    - Sorting each object to its category object classification
    - we classify them based on a pattern
  - Recognize each person's face
    - This might be identification (who is this person?)
    - or authentication (is this Person is who claims to be associated with that face?)



# Another type of PR

- Sometimes the classes are unknown, still we can still divide them into groups based on their characteristics (patterns)
  - This is called clustering

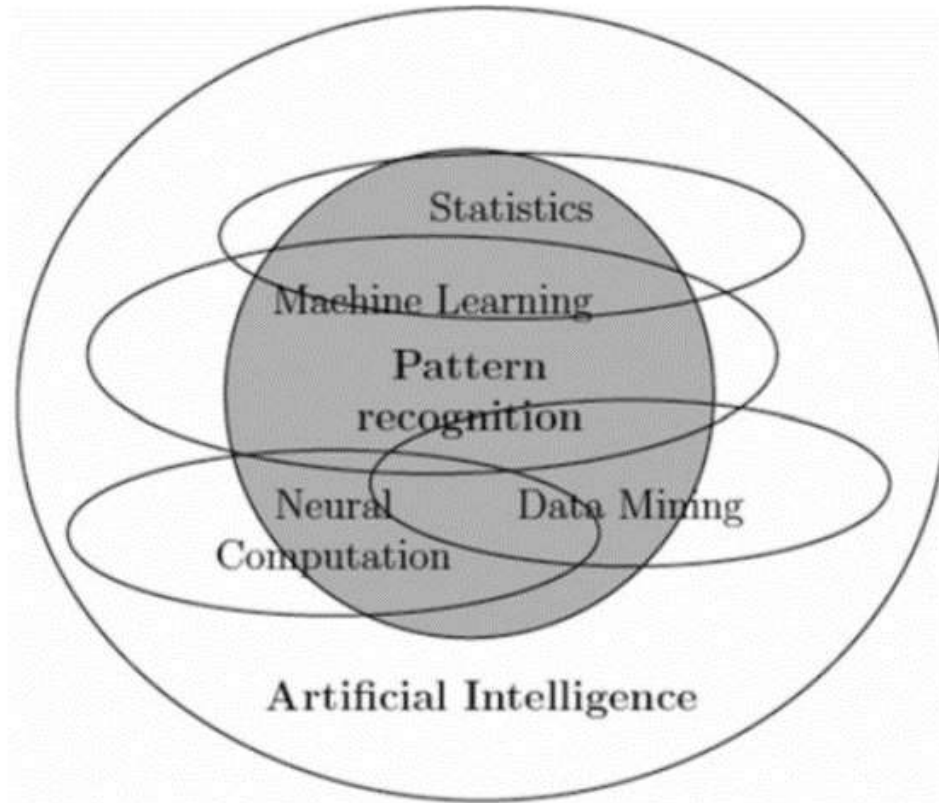




# Definition

- **Pattern Recognition:** Pattern recognition is the process of automatically identifying regularities in data and making inferences or decisions based on those patterns.
    - regularities do not necessarily imply a repetition
  - **A class** is a collection of objects that are similar, but not necessarily identical, and which distinguishable from other classes
- 
- 
- 
- 

# Related fields



- Pattern recognition involves several algorithms from various domains
- Some PR algorithms rely on statistics, while others rely on neural networks and data mining algorithms



# PR applications

- PR has a wide verity of applications and rea-life use cases
  - Interest in pattern recognition and classification has grown due to emerging applications in different domains
  - Some applications are not only challenging but also computationally demanding.
  - These applications include:
- 
- 
- 
- 



# Biometrics

- Biometrics (personal identification based on physical attributes of the face, iris, fingerprints, etc.)





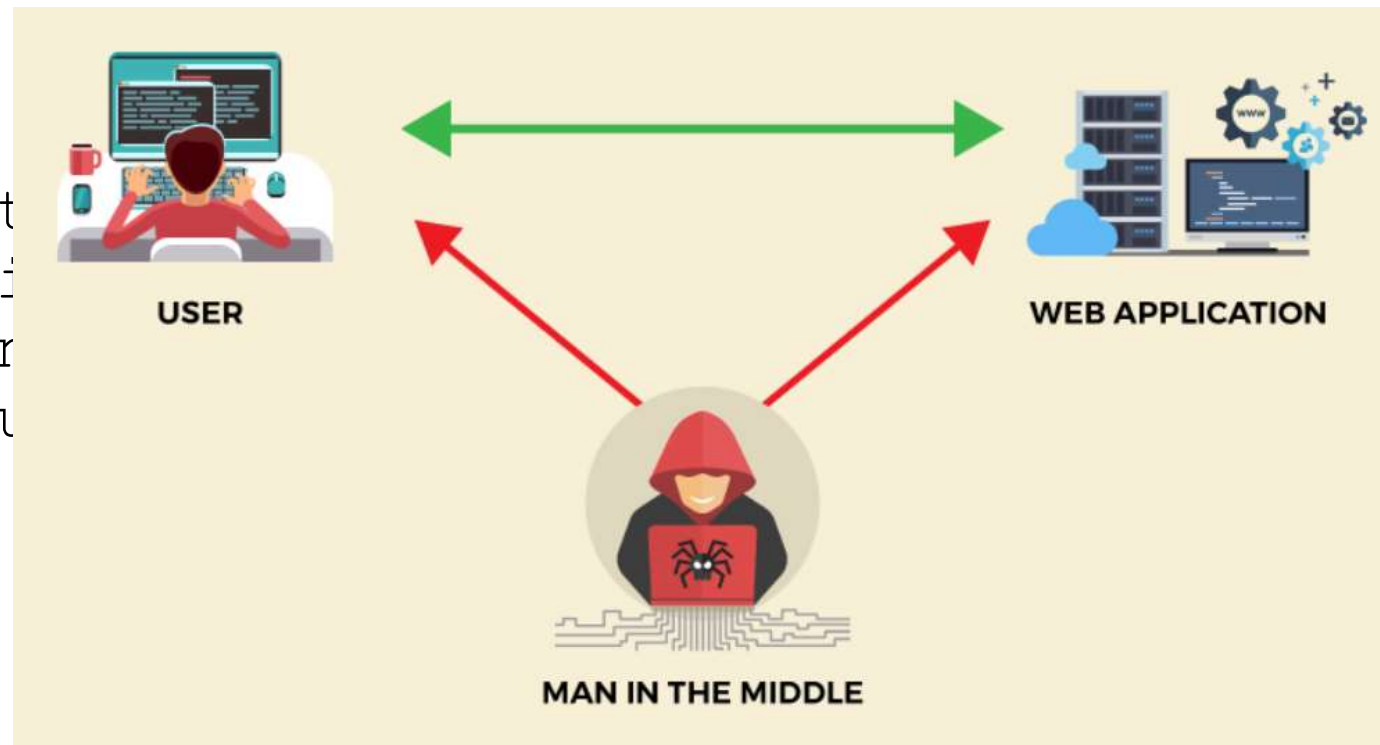
# Machine vision

- Machine vision
- e.g., automated visual inspection in a production line
- Self-driving cars

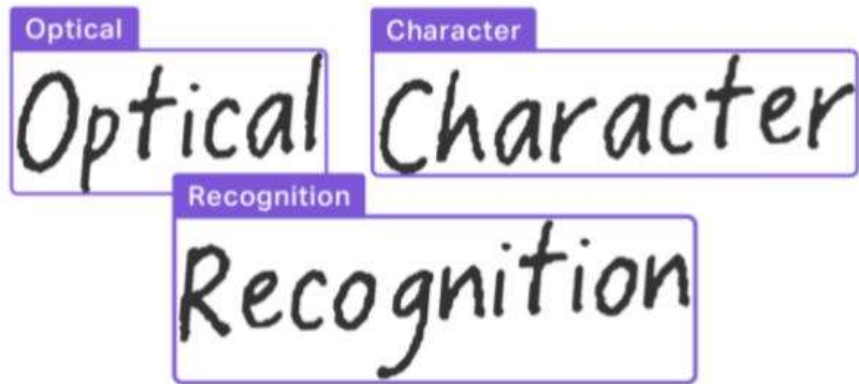


# Network security

- We can find threat patterns, and design a system to prevent any possible intrusion
  - Man in the middle
  - denial of service



# Character recognition

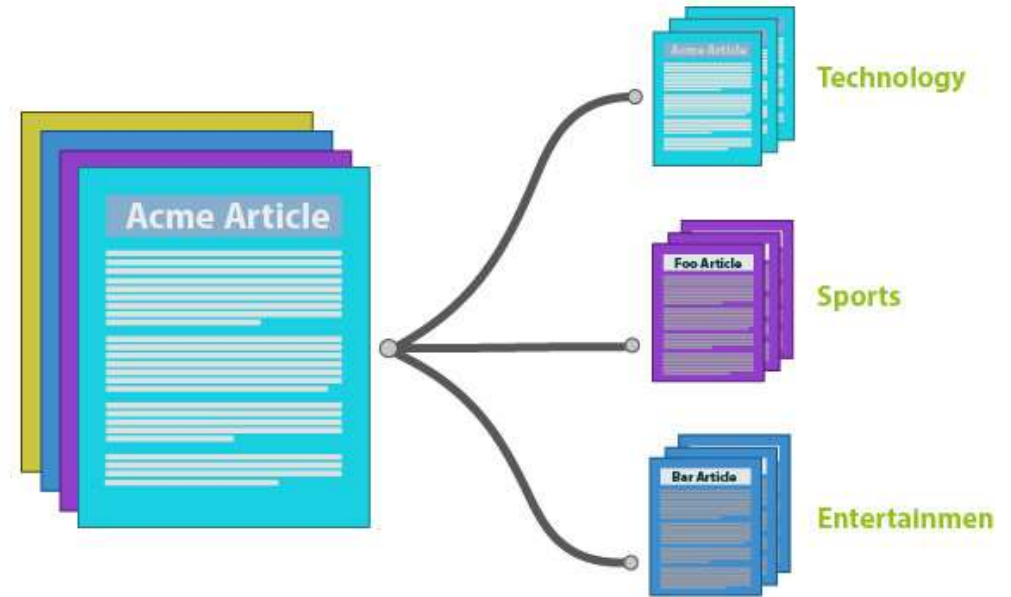


Optical character  
recognition (OCR):

OCR aims to make the  
computer able to read text  
from images

# Document recognition

- Examples:
  - recognize whether an e-mail is spam or not, based on the message header and content
  - classifying invoices into hand-written, machine-printed or receipts







X-ray of a hand, with automatic calculation of bone age by a computer software.

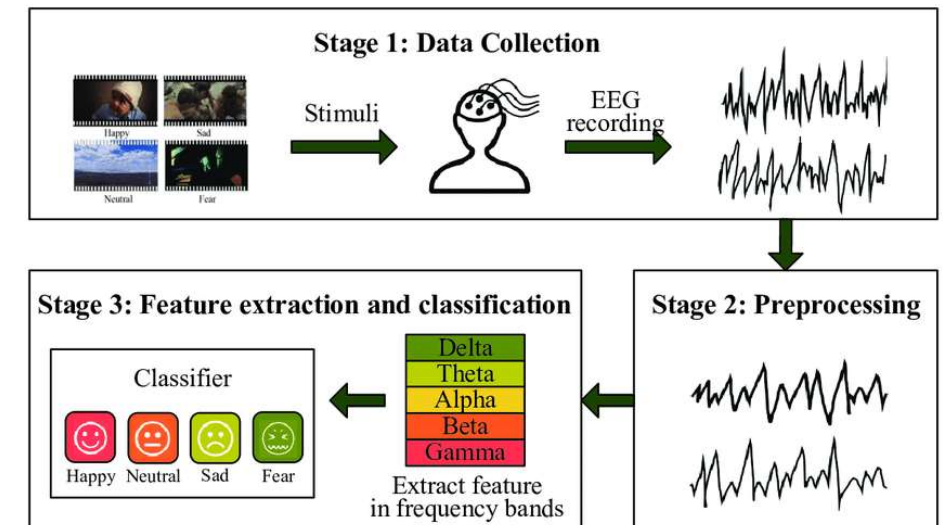
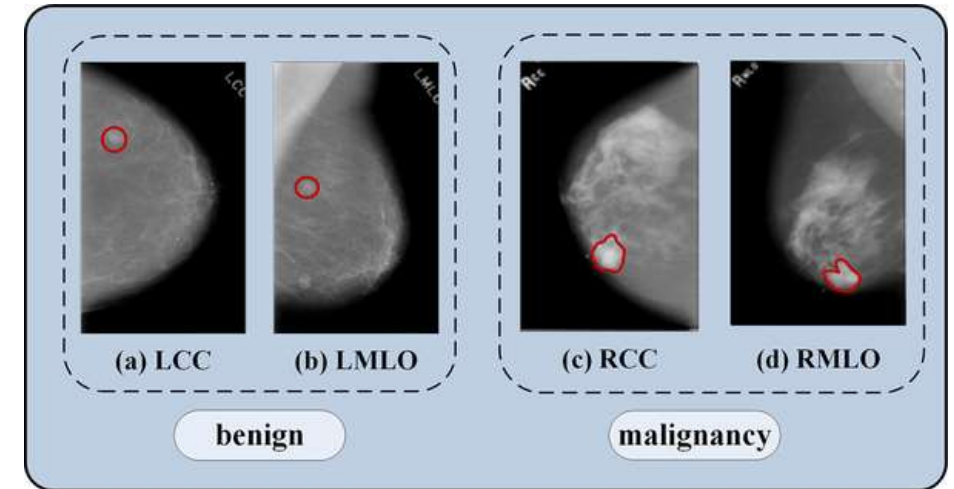
# Computer-aided diagnosis

---

- helping doctors make diagnostic decisions based on interpreting medical data such as mammographic images, ultrasound images

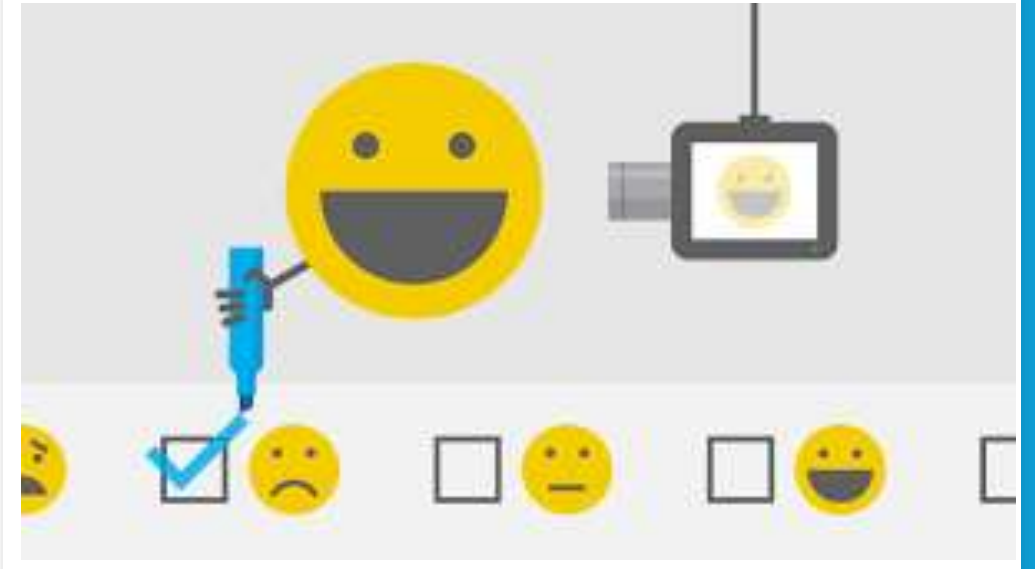
# Medical imaging

- Classifying cells as malignant or benign based on the results of magnetic resonance imaging (MRI) scans
- Classify different emotional and cognitive states from the images of brain activity in functional MRI

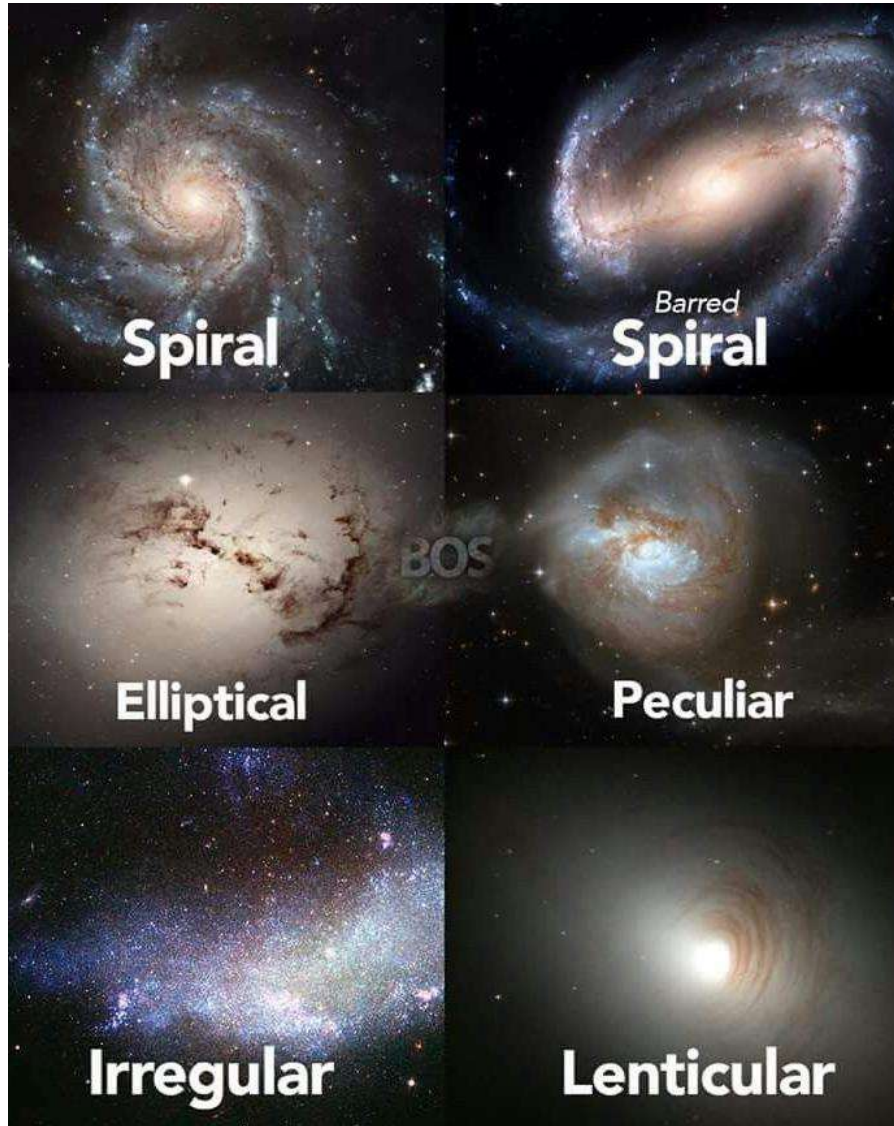


# Social media

- "I spent the whole day looking for a new job, I have not found any job that fits my qualifications, so sad...."
- How this person feels?
- Send targeted ads, job offers, etc.



# Astronomy



- Classifying galaxies based on their shapes
- Automated searches such as the Search for Extra-Terrestrial Intelligence (SETI) which analyzes radio telescope data in an attempt to locate signals that might be artificial in origin (from other planets)

# PR diagram

- A typical PR diagram contains several stages

